

Limits – 9

Sequences defined by integrals and sums

Homework

1 сентября 2026 года

Theoretical minimum. Riemann sums and the error estimate; averaging of fast oscillations and the Riemann–Lebesgue lemma; passage to the limit under the integral sign (monotone and dominated convergence); Laplace’s method; the asymptotics of $\int_0^1 f^n$ and of $\int_0^1 x^n g(x) dx$; the connection with Stirling’s formula.

1. Let g be continuous on $[0, 1]$. Prove that $n \int_0^1 x^n g(x) dx \rightarrow g(1)$, and find the next term of the asymptotics for smooth g .
2. Compute $\lim_{n \rightarrow \infty} \int_0^1 \frac{n dx}{1+n^2 x^2}$ and $\lim_{n \rightarrow \infty} \int_0^1 \frac{nx dx}{1+n^2 x^2}$, explaining why in one of the two cases the limit and the integral do not commute.
3. Let f be continuous on $[0, 1]$. Prove that $\lim_{n \rightarrow \infty} \int_0^1 f(x) \{nx\} dx = \frac{1}{2} \int_0^1 f$, where $\{\cdot\}$ is the fractional part.
4. Prove that $(\int_0^1 f^n)^{1/n} \rightarrow \max_{[0,1]} f$ for a continuous positive f , and refine this: find the asymptotics of $\int_0^1 f^n$ when the maximum is attained at a single non-degenerate interior point.
5. Let $a_n = \int_0^\pi x^n \sin x dx$. Find the asymptotics of a_n and the limit $\frac{a_{n+1}}{na_n}$.
6. Let f be continuous on $[0, \infty)$ with $\int_0^\infty f$ convergent. Prove that $\frac{1}{n} \sum_{k=1}^n f(\frac{k}{n}) \rightarrow \int_0^1 f$, and give an example of an integrable but not Riemann integrable f for which this fails.
7. Prove that $\lim_{n \rightarrow \infty} n \left(\frac{1}{n} \sum_{k=1}^n f(\frac{k}{n}) - \int_0^1 f \right) = \frac{f(1)-f(0)}{2}$ for smooth f , and verify it on $f(x) = x^2$.
8. Let $I_n = \int_0^1 \frac{dx}{1+x^n}$. Find $\lim I_n$ and the asymptotics of $1 - I_n$ as $n \rightarrow \infty$.

9. Prove that $\int_0^1 (1-x^2)^n dx \sim \frac{\sqrt{\pi}}{2\sqrt{n}}$ and deduce an estimate of the volume of the ball in $\mathbb{R}^n$ against the volume of the cube.
10. Let f be continuous and 2π -periodic. Prove that $\frac{1}{n} \sum_{k=1}^n f(k\alpha) \rightarrow \frac{1}{2\pi} \int_0^{2\pi} f$ for irrational $\frac{\alpha}{2\pi}$, and explain how this is related to the sheet on equidistribution in the trigonometry series.